Net force first, then interpret the motion

LEARNING TARGET · R7 I can solve with $F = ma$ — always finding the NET force first.

SECTION A KEY VOCABULARY

Force sum	Add signed forces along the same axis.
Acceleration	Velocity change from net force per mass.
Velocity	Current motion; it can oppose a.
Speed trend	Opposite v and a: slows before reversal.

SECTION B KEY EQUATIONS + UNITS

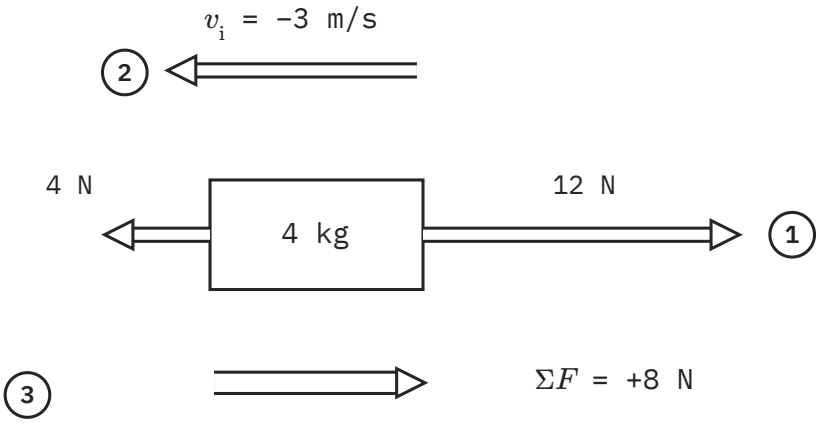
$$\Sigma F_x = ma_x$$

$$v_f = v_i + a_x t$$

ΣF_x	$F_{right} - F_{left}$
+	right

SECTION C COLOR IT · FORCE OPPOSES MOTION

The cart moves left.



after 1 s: v = −1 m/s; speed is smaller

- ☐ **1 · ORANGE** — Forces and separate net
- ☐ **2 · PURPLE** — Leftward velocity
- ☐ **3 · RED** — Slows for the first **1 s**

SECTION D WORKED EXAMPLE

A **4 kg** cart has forces of **12 N** right and **4 N** left. Its initial velocity is **−3 m/s**. **Find a and v after 1 s.**

GIVEN	EQUATION	WORK	ANSWER
m = **4 kg** F_{right} = **12 N**; F_{left} = **4 N** v_i = **−3 m/s**	Sum forces first. $\Sigma F_x = ma_x$. $v = v_i + a_x t$.	ΣF_x = 12 − 4 = **+8 N** a_x = 8/4 = **+2 m/s²** v = −3 + 2(1)	v = **−1 m/s** Still leftward; speed is now **1 m/s**.

SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: FORCE TO MOTION

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else